

ENGR 213: Applied Ordinary Differential Equations

Quizzes & Term Tests Official Solved Examination Bank (Winter 2025)

Concordia University · Gina Cody School of Engineering and Computer Science

Official Document Bank: In-Class Quizzes 1–4 · Term Test 1 (Version 2) · Term Test 2 · 14 Comprehensive Solved Questions

Academic Session: Winter 2025 / Fall 2026 · **Pedagogical Standards:** Full Analytical Proofs & Exact Solutions

ENGR 213: Applied Ordinary Differential Equations

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Department of Mechanical, Industrial & Aerospace Engineering · Concordia University

Academic Session: Winter 2025 / Fall 2026 · **Course:** ENGR 213

Curriculum Mapping: In-Class Quizzes (Quizzes 1–4) · Term Test 1 (Version 2) · Term Test 2 · Comprehensive Step-by-Step Solutions

Executive Summary & Examination Blueprint

This master document compiles and rigorously solves all official in-class quizzes and departmental term tests administered in ENGR 213 during the Winter 2025 academic session. Every problem is numbered sequentially from **Question 1** through **Question 14**, providing verbatim problem statements, classification theorems, step-by-step mathematical derivations, boxed final explicit answers, and professor grading insights.

Summary of Covered Examination Scope

- **Part I: Official In-Class Quizzes (Questions 1–4)**

* **Question 1 (Quiz #1-G · 15 Min):** First-Order Nonlinear Separable Initial Value Problem ($yy' + x^2 = 4, y(0) = 2$).

* **Question 2 (Quiz #2-G · 15 Min):** Physical Modeling of Rates of Change Inversely Proportional to the Square of the Dependent Variable ($dy/dt = k/y^2, y(0) = 1, y(1) = 2$).

* **Question 3 (Quiz #3-G · 15 Min):** Second-Order Constant-Coefficient Nonhomogeneous ODE via the Method of Undetermined Coefficients ($y'' + 5y' + 4y = \cos(3x)$).

* **Question 4 (Quiz #4 · 20 Min):** Power Series Solution about the Ordinary Point $x_0 = 0$ ($y'' - (x + 1)y = 0$, first 6 nonzero terms).

- **Part II: Official Term Test 1 — Version 2 (Questions 5–9 · Feb 21, 2025 · Marked on 100)**

* **Question 5 (Test 1 #1/20):** First-Order ODEs: (1) Direct Integration **Math input error** and (2) First-Order Linear ODE via Integrating Factor **Math input error**.

* **Question 6 (Test 1 #2/15):** Complex Number Arithmetic: Addition, Division, Polar Modulus, and Multiplication ($z_1 = 4 + 5i$, $z_2 = -1 + 2i$).

* **Question 7 (Test 1 #3/20):** Exact Differential Equation IVP:
 $(5y + 3t - 5)dt + (6y + 5t)dy = 0$, $y(-1) = 0$.

* **Question 8 (Test 1 #4/25):** Logistic Differential Equation IVP for Technology Adoption: **Math input error**.

* **Question 9 (Test 1 #5/20):** Forensic Application of Newton's Law of Cooling: Time of Death and Ambient Thermal Decay.

• Part III: Official Term Test 2 (Questions 10–14 · March 27, 2025 · Marked on 100)

* **Question 10 (Test 2 #1/20):** Cauchy-Euler Equidimensional Second-Order ODE: $x^2y'' + 5xy' + 8y = 0$.

* **Question 11 (Test 2 #2/20):** Variation of Parameters for Repeated Root Second-Order ODE:
 $y'' - 4y' + 4y = (x + 1)e^{2x}$.

* **Question 12 (Test 2 #3/20):** Second-Order Homogeneous Initial Value Problem with Critically Damped Roots: $y'' + y' + 0.25y = 0$, $y(0) = 3$, $y'(0) = -3.5$.

* **Question 13 (Test 2 #4/20):** Mechanical Vibrations of a Damped Spring-Mass System ($m = 2\text{extkg}$, $k = 72\text{extN/m}$, $c = 4\text{extN} \cdot \text{exts/m}$).

* **Question 14 (Test 2 #5/20):** Forced Damped Harmonic Oscillator IVP:
 $x'' + 4x' + 4x = 10 \cos(3t)$, $x(0) = 0$, $x'(0) = 0$.

Part I: Official In-Class Quizzes (Winter 2025)

QUIZ

Question 1: First-Order Nonlinear Separable IVP (Official Quiz #1-G · 15 Minutes)

- **Score Weighting:** 10 Marks
- **Problem Statement:** Solve the following equation using a method seen during the lectures. Give your answer in explicit form:

$$yy' + x^2 = 4, \quad y(0) = 2$$

1. Mathematical Classification

This is a first-order, non-linear ordinary differential equation that is separable into the form $g(y) dy = f(x) dx$.

2. Analytical Step-by-Step Derivation

1. ****Isolate the differential terms:****

Express y' as **Math input error**.

Math input error

2. ****Separate the variables:****

Multiply both sides by dx :

$$y dy = (4 - x^2) dx$$

3. ****Integrate both sides:****

$$\int y dy = \int (4 - x^2) dx$$

Math input error

4. ****Solve algebraically for y^2 :**

Multiply the entire relation by 2:

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5. ****Apply the initial condition $y(0) = 2$:**

Math input error

6. ****Formulate the explicit solution:****

Math input error

Taking the square root gives **Math input error**. Because the initial condition requires $y(0) = +2 > 0$, we must select the positive branch:

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3. Verification & Interval of Definition

* Differentiating $y(x)$:

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* The solution remains valid on the open interval containing $x = 0$ where Math input error.

Final Explicit Answer:

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Math input error

QUIZ

Question 2: Nonlinear Rate of Change & Proportionality IVP (Official Quiz #2-G · 15 Minutes)

- **Score Weighting:** 10 Marks

- **Problem Statement:** Consider a process (y) as a function of time (t). This process is known to behave in the following way: The rate of change of y as a function of time is inversely proportional to the square of y .

* (a) Find the expression of the process as a function of time.

* (b) Find the value of y at $t = 5$, if we know $y(0) = 1$ and $y(1) = 2$. Show all your steps.

1. Mathematical Formulation

* The phrase "rate of change of y is inversely proportional to the square of y " translates to:

Math input error

where k is a positive constant of proportionality.

2. Analytical Step-by-Step Derivation

**Part (a): General and Particular Expression $y(t)$ **

1. **Separate variables:**

$$y^2 dy = k dt$$

2. **Integrate both sides:**

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$$y^3 = 3kt + 3C = 3kt + C_1 \quad (\text{extwhere } C_1 = 3C)$$

$$y(t) = \sqrt[3]{3kt + C_1}$$

3. **Determine constant C_1 using initial condition $y(0) = 1$:**

$$1 = \sqrt[3]{3k(0) + C_1} \implies 1 = \sqrt[3]{C_1} \implies C_1 = 1$$

Thus, $y(t) = \sqrt[3]{3kt + 1}$.

4. **Determine the proportionality constant k using condition $y(1) = 2$:**

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5. **Substitute $3k = 7$ back into the expression:**

$$y(t) = \sqrt[3]{7t + 1}$$

**Part (b): Value of Process at $t = 5$ **

Substitute $t = 5$ into the derived function:

Math input error**Final Explicit Answers:**

> (a)

$$y(t) = \sqrt[3]{7t + 1}$$

> (b)

$$y(5) = \sqrt[3]{36}$$

QUIZ

Question 3: Second-Order Nonhomogeneous ODE via Undetermined Coefficients (Official Quiz #3-G · 15 Minutes)

- **Score Weighting:** 10 Marks
- **Problem Statement:** Find the general solution of the equation using the Undetermined Coefficient Technique:

$$y'' + 5y' + 4y = \cos(3x)$$

1. Method Overview

The complete general solution consists of the sum of the complementary homogeneous solution $y_c(x)$ and a particular solution $y_p(x)$:

$$y(x) = y_c(x) + y_p(x)$$

2. Analytical Step-by-Step Derivation

1. ****Find the Complementary Solution $y_c(x)$:****

Set the right-hand side to zero: $y'' + 5y' + 4y = 0$.

Auxiliary / Characteristic equation:

$$m^2 + 5m + 4 = 0 \iff (m + 4)(m + 1) = 0$$

The roots are distinct real numbers: $m_1 = -4$ and $m_2 = -1$.

$$y_c(x) = C_1 e^{-4x} + C_2 e^{-x}$$

2. ****Formulate the Particular Trial Solution $y_p(x)$:****

The driving function is $g(x) = \cos(3x)$. Since $\pm 3i$ is not a root of the characteristic equation, no modification factor is needed.

$$y_p(x) = A \cos(3x) + B \sin(3x)$$

3. ****Compute the derivatives:****

$$y_p'(x) = -3A \sin(3x) + 3B \cos(3x)$$

$$y_p''(x) = -9A \cos(3x) - 9B \sin(3x)$$

4. ****Substitute into the differential operator $L[y] = y'' + 5y' + 4y$:****

$$(-9A \cos(3x) - 9B \sin(3x)) + 5(-3A \sin(3x) + 3B \cos(3x)) + 4(A \cos(3x) + B \sin(3x)) = \cos(3x)$$

Group coefficients of $\cos(3x)$ and $\sin(3x)$:

$$\cos(3x) : (-9A + 15B + 4A) = -5A + 15B = 1 \quad \text{ext} - - [Eq. 1]$$

$$\sin(3x) : (-9B - 15A + 4B) = -15A - 5B = 0 \quad \text{ext} - - [Eq. 2]$$

5. ****Solve the linear algebraic system:****

From [Eq. 2]:

$$-15A = 5B \implies B = -3A$$

Substitute into [Eq. 1]:

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Calculate B :

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Thus, the particular solution is:

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6. **Form the full general solution:**

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Final Explicit Answer:

>

Math input error

QUIZ

Question 4: Power Series Solution about Ordinary Point $x_0 = 0$ (Official Quiz #4 · 20 Minutes)

- **Score Weighting:** 10 Marks
- **Problem Statement:** Using Power Series, find the solution of the equation around the ordinary point 0 (first 6 non-zero terms):

$$y'' - (x + 1)y = 0$$

Steps Required:

1. Trial solution
2. Sum the power series (adjust k in the power and then align summation indices)
3. Find the recurrence relation
4. Write the answer

1. Verification of Ordinary Point

The standard form is $y'' + P(x)y' + Q(x)y = 0$, where $P(x) = 0$ and $Q(x) = -(x + 1)$. Both coefficients are polynomials and analytic everywhere. Hence, $x_0 = 0$ is an ordinary point with an infinite radius of convergence ($R = \infty$).

2. Analytical Step-by-Step Derivation

1. **Trial Series and Derivatives:**

$$y(x) = \sum_{n=0}^{\infty} c_n x^n, \quad y'(x) = \sum_{n=1}^{\infty} n c_n x^{n-1}, \quad y''(x) = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2}$$

2. **Substitute into $y'' - xy - y = 0$.**

$$\sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} - x \sum_{n=0}^{\infty} c_n x^n - \sum_{n=0}^{\infty} c_n x^n = 0$$

Distribute x into the second summation:

$$\sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} - \sum_{n=0}^{\infty} c_n x^{n+1} - \sum_{n=0}^{\infty} c_n x^n = 0$$

3. **Shift Indices to match power x^k .**

* **Term 1:** Let $k = n - 2 \implies n = k + 2$. When $n = 2$, $k = 0$:

$$\sum_{k=0}^{\infty} (k+2)(k+1) c_{k+2} x^k$$

* **Term 2:** Let $k = n + 1 \implies n = k - 1$. When $n = 0$, $k = 1$:

$$\sum_{k=1}^{\infty} c_{k-1} x^k$$

* **Term 3:** Let $k = n$. When $n = 0$, $k = 0$:

$$\sum_{k=0}^{\infty} c_k x^k$$

4. **Align Summation Indices to $k = 1$.

Extract the $k = 0$ terms from the first and third summations:

* For $k = 0$: $(0 + 2)(0 + 1)c_2 - c_0 = 2c_2 - c_0$

Setting the constant term to zero gives:

Math input error

For powers $k \geq 1$, combine the summations:

Extra \left or missing \right

5. **Recurrence Relation:

Setting the coefficient of x^k to zero for all $k \geq 1$:

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6. **Iterate to find higher-order coefficients in terms of c_0 and c_1 .

* ** $k = 1$.

Math input error

* ** $k = 2$.

Math input error

* ** $k = 3$.

Math input error

7. **Assemble the General Solution:

Group terms by the two linearly independent arbitrary constants c_0 and c_1 :

$$y(x) = c_0 y_1(x) + c_1 y_2(x)$$

where:

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Final Explicit Answer:

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Math input error

Part II: Official Term Test 1 — Version 2 (Winter 2025 · Feb 21, 2025)

TERM TEST

Question 5: First-Order Separable & Linear ODEs (Test 1 · Problem #1/20)

- **Score Weighting:** 20 Marks
- **Problem Statement:** Find the general solution of the first-order ODEs below:

1. Math input error

2. Math input error

1. Solution to Sub-problem 1: Math input error

- **Classification:** First-order directly integrable ordinary differential equation.
- **Derivation:**

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- **Boxed Result:**

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2. Solution to Sub-problem 2: Math input error

- **Classification:** First-order linear ODE in standard form $y' + P(x)y = Q(x)$, with Math input error and $Q(x) = x^4$.
- **Derivation:**

1. **Compute the Integrating Factor $I(x)$:**

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For $x > 0$, $I(x) = x^3$.2. **Multiply the differential equation by $I(x)$:**

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3. **Integrate both sides with respect to x :**

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4. **Solve explicitly for $y(x)$:**

Math input error

Final Explicit Answers:

> 1.

Math input error

> 2.

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TERM TEST

Question 6: Complex Numbers Arithmetic & Polar Modulus (Test 1 · Problem #2/15)

- **Score Weighting:** 15 Marks
- **Problem Statement:** Consider the following two complex numbers:

$$z_1 = 4 + 5i \quad \text{extand} \quad z_2 = -1 + 2i$$

- * (a) Find $z_1 + z_2$
- * (b) Find **Math input error** and the modulus of the ratio.
- * (c) Find $z_1 z_2$

Analytical Step-by-Step Derivation

****Part (a): Addition $z_1 + z_2$ ****

$$z_1 + z_2 = (4 + 5i) + (-1 + 2i) = (4 - 1) + (5 + 2)i = 3 + 7i$$

****Part (b): Division **Math input error** and Modulus **Math input error******

1. Multiply numerator and denominator by the complex conjugate **Math input error**:

$$\text{Math input error}$$

Since $i^2 = -1$:

$$\text{Math input error}$$

2. Modulus of the quotient:

Using modulus properties **Math input error**:

$$|z_1| = \sqrt{4^2 + 5^2} = \sqrt{16 + 25} = \sqrt{41}$$

$$|z_2| = \sqrt{(-1)^2 + 2^2} = \sqrt{1 + 4} = \sqrt{5}$$

$$\text{Math input error}$$

Direct Cartesian Check:

$$\text{Math input error}$$

****Part (c): Multiplication $z_1 z_2$ ****

$$z_1 z_2 = (4 + 5i)(-1 + 2i) = 4(-1) + 4(2i) + 5i(-1) + 5i(2i) = -4 + 8i - 5i + 10(-1) = -14 + 3i$$

Final Explicit Answers:

> (a)

$$z_1 + z_2 = 3 + 7i$$

> (b)

Math input error

> (c)

$$z_1 z_2 = -14 + 3i$$

TERM TEST

Question 7: Exact Differential Equation IVP (Test 1 · Problem #3/20)

- **Score Weighting:** 20 Marks
- **Problem Statement:** Find the particular solution of the following IVP:

$$(5y + 3t - 5) dt + (6y + 5t) dy = 0, \quad y(-1) = 0$$

1. Test for Exactness

Let $M(t, y) = 5y + 3t - 5$ and $N(t, y) = 6y + 5t$.

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Since Math input error everywhere on $\mathbb{R}^2$, the differential equation is **exact**.

2. Construction of the Potential Function $\Psi(t, y)$

There exists a potential function $\Psi(t, y)$ such that Math input error and Math input error.

1. **Integrate $M(t, y)$ with respect to t :**

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2. **Differentiate $\Psi(t, y)$ with respect to y and equate to $N(t, y)$:**

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3. **Integrate $h'(y)$:**

$$h(y) = 3y^2$$

4. **General Implicit Solution:**

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3. Apply the Initial Condition $y(-1) = 0$

Substitute $t = -1$ and $y = 0$:

Math input error

Multiply through by 2 to clear fractions:

$$6y^2 + 10ty + 3t^2 - 10t - 13 = 0$$

4. Explicit Formulation

Apply the quadratic formula to solve for y :

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Math input error

To satisfy $y(-1) = 0$:

Math input error

Choosing the negative sign yields **Math input error**.

Final Explicit Answer:

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Math input error

TERM TEST

Question 8: Logistic Differential Equation for Technology Adoption (Test

1 · Problem #4/25)

- **Score Weighting:** 25 Marks
- **Problem Statement:** The number $N(t)$ of stores in Canada using a computerized checkout system is described by the initial value problem:

Math input error

Solve the initial value problem (show all your steps) to get $N(t)$, and find how many stores are expected to use this system for $t = 12$.

1. Identification of Parameters

This is the Verhulst logistic growth model:

Math input error

Here, intrinsic growth rate $r = 1$, and Math input error, meaning the carrying capacity (maximum market saturation) is $K = 2500$ stores.

2. Analytical Step-by-Step Derivation

1. **Separate variables:**

Math input error

2. **Partial Fraction Decomposition:**

Math input error

$$2500 = A(2500 - N) + BN$$

Setting $N = 0 \implies A = 1$. Setting $N = 2500 \implies B = 1$.

Math input error

3. **Integrate both sides:**

Math input error

Math input error

4. **Determine C_0 using $N(0) = 2$:

Math input error

5. **Solve explicitly for $N(t)$:

Math input error

Math input error

3. Evaluation at $t = 12$

Substitute $t = 12$:

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Compute e^{-12} :

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Math input error

Math input error

Final Explicit Answers:

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Math input error

> At $t = 12$:

Math input error

TERM TEST Question 9: Forensic Newton's Law of Cooling (Test 1 · Problem #5/20)

- **Score Weighting:** 20 Marks
- **Problem Statement:** A dead body was found in a closed room having a constant temperature of 20°extC . At the time of discovery, the body temperature was 27°extC . One hour later, it was at 24°extC . Assume the time of death is $t = 0$ and that a living body has a temperature of 37°extC . How many hours elapsed before the body was found?

*(Hint: use Newton's law of cooling: **Math input error**)*

1. Mathematical Formulation

* Ambient temperature: $T_m = 20^{\circ}\text{extC}$.

* Initial body temperature at death ($t = 0$): $T(0) = 37^{\circ}\text{extC}$.

* Discovery occurs at an unknown elapsed time t_d . Measured: $T(t_d) = 27^{\circ}\text{extC}$.

* One hour post-discovery ($t = t_d + 1$): $T(t_d + 1) = 24^{\circ}\text{extC}$.

2. Analytical Step-by-Step Derivation

1. **Solve Newton's Law of Cooling:**

Math input error

$$\ln|T - 20| = kt + C \implies T(t) - 20 = C_0 e^{kt} \implies T(t) = 20 + C_0 e^{kt}$$

2. **Apply Initial Condition $T(0) = 37$:

$$37 = 20 + C_0 e^0 \implies C_0 = 17$$

Therefore, the temperature trajectory is:

$$T(t) = 20 + 17e^{kt}$$

3. **Formulate the system of two observation equations:**

* At discovery $t = t_d$:

Math input error

* One hour later $t = t_d + 1$:

$$24 = 20 + 17e^{k(t_d+1)} \implies 17e^{kt_d}e^k = 4 \quad \text{ext} - - - [\text{Eq. B}]$$

4. **Determine the cooling constant k :

Substitute [Eq. A] into [Eq. B]:

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Math input error

5. **Calculate the elapsed time t_d .

From [Eq. A], take the natural logarithm:

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Evaluate numerical values:

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Math input error

Convert to hours and minutes:

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Final Explicit Answer:

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Math input error

Part III: Official Term Test 2 (Winter 2025 · March 27, 2025)

TERM TEST

Question 10: Cauchy-Euler Equidimensional ODE (Test 2 · Problem #1/20)

- **Score Weighting:** 20 Marks
- **Problem Statement:** Find the general solution of the ODE below:

$$x^2 y'' + 5xy' + 8y = 0 \quad (x > 0)$$

1. Mathematical Classification

This is a second-order homogeneous Cauchy-Euler (equidimensional) differential equation with constant coefficients $a = 1$, $b = 5$, and $c = 8$.

2. Analytical Step-by-Step Derivation

1. **Trial Substitution:**

Assume a solution of the form $y(x) = x^m$.

$$y' = mx^{m-1}, \quad y'' = m(m-1)x^{m-2}$$

2. **Substitute into the differential equation:**

$$\begin{aligned} x^2[m(m-1)x^{m-2}] + 5x[mx^{m-1}] + 8[x^m] &= 0 \\ x^m[m(m-1) + 5m + 8] &= 0 \end{aligned}$$

3. **Form and solve the auxiliary characteristic equation:**

$$m^2 - m + 5m + 8 = 0 \iff m^2 + 4m + 8 = 0$$

Use the quadratic formula:

Math input error

The roots are complex conjugates: Math input error and Math input error.

4. **General Solution Formula for Complex Conjugate Roots:**

For Cauchy-Euler equations with roots Math input error:

Math input error

Substitute Math input error and Math input error:

Math input error

Final Explicit Answer:

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Extra \left or missing \right

TERM TEST**Question 11: Variation of Parameters for 2nd-Order ODE (Test 2 · Problem #2/20)**

- **Score Weighting:** 20 Marks
- **Problem Statement:** Using Variation of parameter technique, solve:

$$y'' - 4y' + 4y = (x + 1)e^{2x}$$

1. Homogeneous Solution and Basis Functions

Auxiliary equation:

$$m^2 - 4m + 4 = 0 \iff (m - 2)^2 = 0 \implies m_1 = m_2 = 2 \text{ (repeated real root)}$$

Fundamental set of solutions:

$$y_1(x) = e^{2x}, \quad y_2(x) = xe^{2x}$$

$$y_c(x) = C_1 e^{2x} + C_2 x e^{2x}$$

2. Wronskian Determinant Calculation

$$W(y_1, y_2)(x) = \begin{vmatrix} e^{2x} & x e^{2x} \\ 2e^{2x} & (1 + 2x)e^{2x} \end{vmatrix} = e^{4x} \\ (1 + 2x)e^{2x} - 2x e^{2x} e^{2x} = e^{4x}(1 + 2x - 2x) = e^{4x}$$

eq 0

3. Variation of Parameters Formula

With driving term $g(x) = (x + 1)e^{2x}$, particular solution is $y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$:

1. Calculate $u_1(x)$:

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Math input error

2. Calculate $u_2(x)$:

Math input error

Math input error

3. Assemble and Simplify $y_p(x)$:

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Combine like terms:

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4. General Solution:

Math input error

Final Explicit Answer:

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Math input error

TERM TEST**Question 12: Second-Order Homogeneous IVP with Repeated Roots (Test 2 · Problem #3/20)**

- **Score Weighting:** 20 Marks
- **Problem Statement:** Find the particular solution of the following IVP:

$$y'' + y' + 0.25y = 0, \quad y(0) = 3, \quad y'(0) = -3.5$$

1. Analytical Step-by-Step Derivation

1. **Auxiliary Equation:**

Math input error

Repeated real root: Math input error.

2. **General Solution:**

$$y(x) = (C_1 + C_2x)e^{-0.5x}$$

3. **Compute the Derivative $y'(x)$:**

Extra \left or missing \right

4. **Apply Initial Conditions:**

* $y(0) = 3$:

$$(C_1 + C_2(0))e^0 = 3 \implies C_1 = 3$$

* $y'(0) = -3.5$:

$$C_2 - 0.5(3) = -3.5 \implies C_2 - 1.5 = -3.5 \implies C_2 = -2$$

5. **Formulate the Particular Solution:**

$$y(x) = (3 - 2x)e^{-0.5x}$$

Final Explicit Answer:

>

$$y(x) = (3 - 2x)e^{-0.5x}$$

TERM TEST

Question 13: Mechanical Vibrations of Damped Spring-Mass System (Test 2 · Problem #4/20)

- **Score Weighting:** 20 Marks
 - **Problem Statement:** A 2 kg mass is attached to a spring of constant 72 N/m. The system is horizontal and the ground provides a damping force equal to 4 times the velocity.
- * (a) Write the equation for the forces involved, starting with Newton's second law (sum of forces).
- * (b) Solve the equation found in (a).
- * (c) Find the velocity of the mass 1 s after release if the mass is released from a position $x(0) = 2 \text{ m}$ (released from rest, $x'(0) = 0$).

Analytical Step-by-Step Derivation

****Part (a): Newton's Second Law Force Balance****

1. System parameters:

* Mass: $m = 2 \text{ kg}$

* Damping coefficient: $c = 4 \text{ N} \cdot \text{s/m}$ (damping force opposes velocity: **Math input error**)

* Spring constant: $k = 72 \text{ N/m}$ (restoring force opposes displacement: $F_s = -kx$)

2. Sum of forces:

Math input error

$$mx'' + cx' + kx = 0 \implies 2x'' + 4x' + 72x = 0$$

Dividing by 2 gives the standard monic form:

$$x'' + 2x' + 36x = 0$$

****Part (b): Solution of the Homogeneous ODE****

1. Characteristic equation:

Math input error

2. Underdamped motion with decay factor **Math input error** and damped natural frequency **Math input error**:

Extra \left or missing \right

****Part (c): Determination of Velocity at $t = 1 \text{ s}$ ****

1. Apply initial conditions $x(0) = 2 \text{ m}$ and $x'(0) = 0$:

$$x(0) = C_1 = 2$$

2. Differentiate $x(t)$ using the product rule:

$$x'(t) = -e^{-t}[C_1 \cos(\sqrt{35}t) + C_2 \sin(\sqrt{35}t)] + e^{-t}[-\sqrt{35}C_1 \sin(\sqrt{35}t) + \sqrt{35}C_2 \cos(\sqrt{35}t)]$$

Extra \left or missing \right

3. Apply $x'(0) = 0$:

Math input error

4. Simplified velocity function $v(t) = x'(t)$:

Since $\sqrt{35}C_2 - C_1 = 0$, the cosine component vanishes completely:

Extra \left or missing \right

Substitute $C_1 = 2$ and Math input error:

Math input error

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5. Evaluate at $t = 1 \text{ exts}$:

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Numerical values:

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Final Explicit Answers:

> (a)

$$2x'' + 4x' + 72x = 0 \iff x'' + 2x' + 36x = 0$$

> (b)

Extra \left or missing \right

> (c)

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TERM TEST

Question 14: Forced Harmonic Oscillator with Damping IVP (Test 2 · Problem #5/20)

- **Score Weighting:** 20 Marks
- **Problem Statement:** A force is driving a system of mass and spring such as:

$$x'' + 4x' + 4x = 10 \cos(3t)$$

Find the equation of motion if the initial position and velocity are 0 ($x(0) = 0, x'(0) = 0$).

Analytical Step-by-Step Derivation

1. **Homogeneous Complementary Solution $x_h(t)$:**

Characteristic equation:

$$m^2 + 4m + 4 = 0 \iff (m + 2)^2 = 0 \implies m_1 = m_2 = -2 \text{ ext(critically damped)}$$

$$x_h(t) = (C_1 + C_2 t)e^{-2t}$$

2. **Particular Solution $x_p(t)$ via Undetermined Coefficients:**

Driving frequency is $\omega = 3$. Assume:

$$x_p(t) = A \cos(3t) + B \sin(3t)$$

Compute derivatives:

$$x_p'(t) = -3A \sin(3t) + 3B \cos(3t)$$

$$x_p''(t) = -9A \cos(3t) - 9B \sin(3t)$$

Substitute into $x'' + 4x' + 4x = 10 \cos(3t)$:

$$(-9A \cos(3t) - 9B \sin(3t)) + 4(-3A \sin(3t) + 3B \cos(3t)) + 4(A \cos(3t) + B \sin(3t)) = 10 \cos(3t)$$

Collect terms for $\cos(3t)$ and $\sin(3t)$:

$$\cos(3t) : (-9A + 12B + 4A) = -5A + 12B = 10 \quad \text{ext} - - [Eq. 1]$$

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3. **Solve for A and B :**

Substitute [Eq. 2] into [Eq. 1]:

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Compute B :

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Therefore:

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4. **General Solution:**

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5. **Apply Initial Conditions $x(0) = 0$ and $x'(0) = 0$:

* $x(0) = 0$:

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* Differentiate $x(t)$:

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* $x'(0) = 0$:

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6. **Final Equation of Motion:**

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Final Explicit Answer:

>

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